EMBERFALL HATCHERY

Where biology meets myth

https://emberfallhatchery.streamlit.app/

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Written Summary

BIOL 4660 - Biomedical Science Tecniques

From the Desk of Dr. Elara Voss
Lead Geneticist, Emberfall Hatchery

To the apprentice,

By the time you read this, I will have already departed on a short expedition beyond the eastern ranges. A number of unusual specimens have been reported, and I intend to assess their potential for future work at the Hatchery.

In my absence, I have left you a selection of client commissions currently awaiting preliminary design. These are not yet approved for implementation, so there is no immediate pressure. Every construct will be reviewed upon my return before any work proceeds beyond the design stage.

Think of this as an opportunity. Each client presents a different biological challenge. Your task is to interpret their request, construct an appropriate genetic system, and ensure that it behaves exactly as intended. Precision is the difference between an elegant solution and a dangerous one.

You may notice that some components are available that are not strictly necessary. This is intentional. A skilled designer must be able to distinguish between what is possible and what is appropriate.

Complete these assignments before I return. I will review your designs personally.

— Dr. Elara Voss

About Emberfall Hatchery

Emberfall Hatchery is an interactive simulation designed to teach core concepts in genetic engineering through a narrative-driven experience.

Players take on the role of an apprentice genetic designer, building biological systems to meet specific functional constraints. Each scenario introduces real-world principles, including:

- tissue-specific gene expression
- inducible and conditional systems
- transcriptional control (GAL4/UAS, GAL80)
- recombinase-based systems (Cre-lox, FLP-FRT)

Rather than focusing on a single correct answer, the simulation evaluates designs based on how well they satisfy biological requirements, encouraging problem-solving and critical thinking.

Understanding Your Work at Emberfall

Each commission follows a structured process:

1. **Read the Client Letter**: Clients describe their needs in practical or narrative terms. Pay close attention to constraints and intended outcomes.

2. **Review the Scientist's Notes**: These translate the request into biological requirements. This is where the true problem is defined.

3. **Design Your Construct**: Use the available components to build a system that meets all conditions:
   - correct location of expression
   - correct timing of activation
   - avoidance of unintended effects

4. **Submit for Evaluation**: Your design will be analyzed based on how well it satisfies the biological constraints of the scenario.

There is rarely a single "correct" solution but there are many incorrect ones.

The Making of Emberfall Hatchery

The project was developed using Python and Streamlit, with a modular system for:

- scenario-based gameplay
- dynamic component selection
- constraint-based grading
- interactive construct assembly

Source code and implementation details can be found here:

- https://github.com/abk9zz/labnoob.git

Start Your Work at the Hatchery!

- https://emberfallhatchery.streamlit.app/